Solute Redistribution in Solidification

Technical Summary

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1. Scope and material data

A one-dimensional bar of nominal composition $C_0 = 4.8\,\text{wt.\%}$ Cu in Al is solidified from one end. At solid fraction f_s the interface is planar and local equilibrium holds. Four limiting models are compared: equilibrium (lever rule), Scheil–Gulliver, no solid diffusion with limited liquid diffusion, and Clyne–Kurz with an optional Won–Thomas coarsening correction. A temperature slider adds a sensitivity study: it moves the Al–Cu tie-line and rescales k, D_S and D_L from quadratic phase-diagram fits and Arrhenius diffusion.

Quantity	Value	Source / justification
C_0	4.8 wt.% Cu	nominal alloy; MSE 670 notes, pp. 1–3
k	0.173 at 648 °C	Al–Cu phase diagram; notes
T_m	660.3 °C	melting point of pure Al
T_{eut}	548 °C	Al–Cu eutectic / solidus
$D_{S,\text{ref}}$	4.8×10^{-15} m^2/s	notes and standard Al–Cu data
$D_{L,\text{ref}}$	2.4×10^{-9} m^2/s	notes and standard Al–Cu data
v	20 μm/s	process input
t_f	100 s	local solidification time
λ_2	20 μm	secondary dendrite-arm spacing
α_C	0.10	Won–Thomas coarsening constant
L	1 mm	length scale in δ/L
$Q_S,\ Q_L$	130, 30 kJ/mol	Arrhenius activation energies
T_{ref}	648 °C	coupling anchor

Table 1: Reference material and process data.

The tie-line fits use the Al–Cu anchors $(0, 660.3)$, $(4.8, 648)$, $(33, 548)$ for the liquidus and $(0, 660.3)$, $(0.8304, 648)$, $(5.65, 548)$ for the solidus, with compositions in wt.% and temperatures in °C. Values are standard Al–Cu approximations used in the MSE 670 notes and in Kurz and Fisher; they are not a CALPHAD assessment.

2. Assumptions

Common to all four models.

- One-dimensional growth with a planar interface advancing at constant velocity; constant density and no convection.
- Local equilibrium at the interface, $C_S^* = k\,C_L^*$.
- Fick diffusion with one diffusivity per phase.
- k is constant within one model evaluation. The temperature coupling deliberately relaxes this as a sensitivity study.
- The terminal eutectic reaction is not modelled. The 33 wt.% point is used only as a phase-diagram anchor.

Model-specific.

- *Equilibrium:* infinite diffusion in both phases; both phases stay uniform.

- *Scheil–Gulliver:* $D_S = 0$ and the liquid is perfectly mixed.
- *Limited liquid:* $D_S = 0$ and no liquid mixing; a steady diffusion boundary layer forms ahead of the interface. Initial and final transients are captured by a mass-conserving sweep, but the interface is still assumed planar.
- *Clyne–Kurz:* finite back diffusion in the solid with a mixed liquid. It is an analytical approximation and does not resolve a spatial profile. The Won–Thomas correction is approximate; its empirical arm-spacing fits are steel-specific and are not used.

3. Equations

3.1 Interface condition and solute balance

$$C_S^* = k\, C_L^*, \qquad f_s\, \overline{C_S} + (1 - f_s)\, \overline{C_L} = C_0. \tag{1}$$

3.2 Equilibrium (lever rule)

$$C_S = \frac{kC_0}{1 - (1 - k)f_s}, \qquad C_L = \frac{C_0}{1 - (1 - k)f_s}. \tag{2}$$

3.3 Scheil–Gulliver

$$C_L = C_0(1 - f_s)^{k-1}, \qquad C_S^* = kC_L, \qquad \overline{C_S} = \frac{C_0\left[1 - (1 - f_s)^k\right]}{f_s}. \tag{3}$$

For $k < 1$ both interface compositions diverge as $f_s \to 1$; the apps cap the plotted range and flag the singularity.

3.4 No solid diffusion, limited liquid diffusion

Ahead of a planar interface moving at v, the steady boundary layer is

$$C_L(z) = C_0 + (C_L^* - C_0)\, e^{-z/\delta}, \qquad \frac{\delta}{L} = \frac{D_L}{vL}. \tag{4}$$

At steady state $C_L^* = C_0/k$ and the solid grows at $C_S = C_0$. Over the whole bar the implementation integrates the normalised solid profile $S(x) = C_S(x)/C_0$ from local equilibrium plus global conservation. With $I(f_s) = \int_0^{f_s} S\, dx$,

$$\frac{C_L^*}{C_0} = 1 + \frac{f_s - I(f_s)}{\delta\left(1 - e^{-(1-f_s)/\delta}\right)}, \qquad S(f_s) = k\,\frac{C_L^*}{C_0}. \tag{5}$$

Equation (5) is solved on a uniform grid by the trapezoidal rule with eight fixed-point iterations per node ($N = 1400$ intervals in MATLAB, $N = 2200$ in the browser). This reproduces the initial transient, the $C_S = C_0$ plateau and the final spike.

3.5 Clyne–Kurz

$$\alpha = \frac{4D_S t_f}{\lambda_2^2}, \qquad \beta = 2\alpha\left(1 - e^{-1/\alpha}\right) - e^{-1/(2\alpha)}, \tag{6}$$

$$\frac{C_L}{C_0} = \left[1 - (1 - \beta k) f_s\right]^{\frac{k-1}{1-\beta k}}, \qquad C_S^* = kC_L, \qquad \overline{C_S} = \frac{C_0 - (1 - f_s)C_L}{f_s}. \tag{7}$$

The limits are $\beta \to 0$ (Scheil) and $\beta \to 1$ (equilibrium). The cases $k = 1$ and $1 - \beta k = 0$ are evaluated separately to avoid 0/0.

3.6 Won–Thomas coarsening correction

$$\alpha^{+} = 2\,(\alpha + \alpha_C)\,, \tag{8}$$

after which Eq. (6) is evaluated with α^{+}.

3.7 Temperature coupling

The liquidus and solidus fits are

$$T = -0.0298060391\,C_L^2 - 2.41943101\,C_L + 660.3, \tag{9}$$

$$T = -1.05070285\,C_S^2 - 13.93963508\,C_S + 660.3, \tag{10}$$

and the physical root is $C(T) = (-b - \sqrt{b^2 - 4a(c-T)})/(2a)$ with $a < 0$, $b < 0$, $c = 660.3$. The partition coefficient follows the tie-line:

$$k(T) = \frac{C_S(T)}{C_L(T)}, \tag{11}$$

with $k = k_{\mathrm{ref}}$ when $C_L < 10^{-8}$. Both diffusivities use Arrhenius scaling anchored at $T_{\mathrm{ref}} = 648\,^{\circ}\mathrm{C}$:

$$D(T) = D_{\mathrm{ref}} \exp\!\left[-\frac{Q}{R}\left(\frac{1}{T + 273.15} - \frac{1}{T_{\mathrm{ref}} + 273.15}\right)\right]. \tag{12}$$

4. Results and discussion

Table 2 compares the models at two solid fractions. Every entry satisfies the solute balance in Eq. (1) to displayed precision.

Model	$C_S(0.5)$	$C_L(0.5)$	$C_S(0.9)$	$C_L(0.9)$
Equilibrium	1.42	8.18	3.25	18.77
Scheil–Gulliver	1.47	8.52	5.58	32.23
Clyne–Kurz	1.47	8.51	5.53	31.94
Won–Thomas	1.45	8.40	4.37	25.24
Limited liquid	2.90	16.75	5.54	32.01

Table 2: Interface compositions in wt.% at $f_s = 0.5$ and 0.9 for the reference data of Table 1.

At the default state ($f_s = 0.5$) the lever rule gives $C_S = 1.416$ and $C_L = 8.184\,\mathrm{wt.\%}$, the lowest liquid enrichment of the four models. Scheil and Clyne–Kurz are almost identical because the default Fourier number is small, $\alpha = 0.0048$, giving $\beta = 0.0096$; the solid therefore behaves nearly as if it did not diffuse. The Won–Thomas correction raises these to $\alpha^{+} = 0.2096$ and $\beta = 0.3236$, and its curve moves visibly toward the equilibrium limit. Clyne–Kurz thus brackets the two classical limits through β, which is its main pedagogical value.

The limited-liquid model is the most strongly segregating case, especially at intermediate fractions: at $f_s = 0.5$ its liquid reaches 16.75 wt.% versus 8.18 wt.% for equilibrium, because rejected solute has no convective or diffusive escape except the thin boundary layer. Its solid profile shows the expected sequence of an initial transient above kC_0, a plateau near $C_S = C_0$, and a final spike as the boundary layer is squeezed against the mould wall. The default boundary-layer thickness is $\delta/L = D_L/(vL) = 0.12$. Near $f_s = 1$ the Scheil and limited-liquid curves both diverge; the apps stop drawing or flag the singularity rather than inventing a finite value.

Temperature acts mainly through diffusion, not through k (Table 3). Across 548–660 °C the partition coefficient changes only from 0.1712 to 0.1735 (about 1.3%), so the normalised equilibrium and Scheil

curves are almost temperature-independent. In contrast D_S rises by a factor of about 7.9 between 548 and 648 °C, which increases back diffusion and lowers the Clyne–Kurz liquid composition; D_L changes by about 1.6 over the same interval, which changes δ/L and therefore the limited-liquid profile. The temperature slider is therefore best read as a diffusion sensitivity control, not as a phase-diagram prediction.

T (°C)	C_L (wt.%)	C_S (wt.%)	$k(T)$	D_S (m^2/s)	D_L (m^2/s)
548	33.00	5.650	0.1712	6.07×10^{-16}	1.49×10^{-9}
600	20.00	3.436	0.1718	1.89×10^{-15}	1.94×10^{-9}
640	7.67	1.324	0.1727	4.14×10^{-15}	2.32×10^{-9}
648	4.80	0.830	0.1730	4.80×10^{-15}	2.40×10^{-9}
660	0.12	0.021	0.1735	5.97×10^{-15}	2.52×10^{-9}

Table 3: Tie-line and transport properties produced by the temperature coupling.

Numerically, the stand-alone MATLAB functions and the browser implementation agree at the reference points: at 640 °C both give $k = 0.1727$, $D_S = 4.137 \times 10^{-15}$ m^2/s and $D_L = 2.319 \times 10^{-9}$ m^2/s. The limited-liquid grid is converged at the stated N; doubling it changes the plotted profiles by less than the line width. The Scheil average uses its analytic $f_s \to 0$ limit, and the Clyne–Kurz special cases avoid 0/0.

5. Limitations

The quadratic fits are extrapolated over 548–700 °C and should be used near the alloy composition, not as a full phase-diagram prediction. Constant k, constant density, a planar interface and the neglect of the eutectic reaction are the main physical restrictions. The Clyne–Kurz and Won–Thomas forms are analytical approximations, and the limited-liquid model assumes a steady boundary layer even though the concentration sweep captures the end transients. The notes' spatial model with a thermal gradient G_L and the constitutional-supercooling criterion is not implemented: it needs a distance axis and a G_L input that the current app does not have. The results should therefore be read as a consistent teaching comparison of limiting cases, not as a casting simulation.